

# Splunk Introduction - Analysis of a Log

This assignment involved analyzing a sample system log file using Splunk to practice log ingestion, search, reporting, and dashboard creation. The objective was to develop foundational skills in security log analysis and data visualization using Splunk's powerful search capabilities.

The log file `system_log_example_1000.txt` was uploaded via Splunk Web using the **Add Data** feature. The source type was set to `syslog` and the data was indexed under a custom index named `system_logs_1`.

The screenshot displays the Splunk Search interface. The search bar contains the query `index=system_logs_1`. Below the search bar, it indicates that 1,000 events were found. The interface shows a timeline visualization at the top and a list of search results below. The search results table includes columns for Time and Event. The events listed are:

| Time                    | Event                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023-01-01 23:59:42.000 | ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs |
| 2023-01-01 23:58:34.000 | ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs |
| 2023-01-01 23:58:08.000 | WARN User login attempt failed<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                     |
| 2023-01-01 23:57:48.000 | WARN Service response time is slow<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                 |
| 2023-01-01 23:57:47.000 | INFO Service started successfully<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                  |
| 2023-01-01 23:56:49.000 | ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs |
| 2023-01-01 23:56:04.000 | WARN User login attempt failed<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                     |

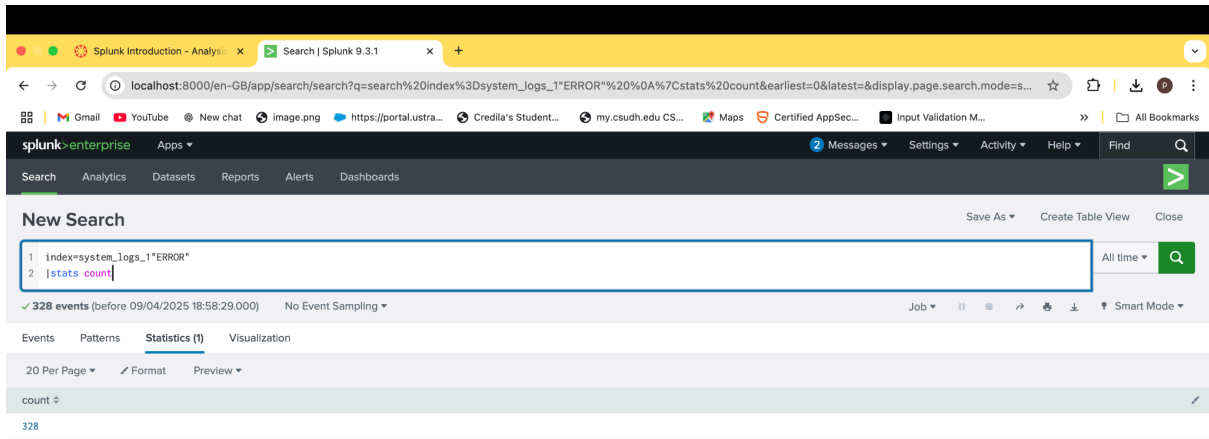
## Searching and Analyzing Logs

The log data was explored using the **Search & Reporting** app. Key search queries included:

- Displaying all logs from the file.
- Filtering and counting log levels such as "ERROR", "INFO", and "WARN".
- Identifying top sources or hosts.

- Viewing logs by time and event frequency.

Splunk's SPL (Search Processing Language) was used to extract meaningful insights from the dataset.



Splunk Introduction - Analysis | Search | Splunk 9.3.1

localhost:8000/en-GB/app/search/search?q=search%20index%3Dsystem\_logs\_1\*ERROR\*&earliest=0&latest=&display.page.search.mode=smart&dispatch.sample\_ratio=... All time

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards

### New Search

Save As Create Table View Close

1 index=system\_logs\_1\*ERROR\* All time

✓ 328 events (before 09/04/2025 18:57:54.000) No Event Sampling Job Smart Mode

Events (328) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 20 Per Page

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- # date\_hour 24
- # date\_mday 1
- # date\_minute 59
- # date\_month 1
- # date\_second 59
- # date\_vday 1
- # date\_year 1
- # date\_zone 1
- a index 1
- # linecount 1
- a punct 3
- a splunk\_server 1
- # timeendpos 1

| i | Time                    | Event                                                                                                                                                                                   |
|---|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 01/01/2023 23:59:42.000 | 2023-01-01 23:59:42 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local source = system_log_example_1000 (4).txt sourcetype = Analysis of logs |
| > | 01/01/2023 23:58:34.000 | 2023-01-01 23:58:34 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local source = system_log_example_1000 (4).txt sourcetype = Analysis of logs |
| > | 01/01/2023 23:56:49.000 | 2023-01-01 23:56:49 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local source = system_log_example_1000 (4).txt sourcetype = Analysis of logs |
| > | 01/01/2023 23:55:43.000 | 2023-01-01 23:55:43 ERROR Service failed to start<br>host = Pradeepreddys-MacBook-Pro.local source = system_log_example_1000 (4).txt sourcetype = Analysis of logs                      |
| > | 01/01/2023 23:40:04.000 | 2023-01-01 23:40:04 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local source = system_log_example_1000 (4).txt sourcetype = Analysis of logs |
| > | 01/01/2023 23:38:30.000 | 2023-01-01 23:38:30 ERROR User login failed<br>host = Pradeepreddys-MacBook-Pro.local source = system_log_example_1000 (4).txt sourcetype = Analysis of logs                            |
| > | 01/01/2023 23:38:13.000 | 2023-01-01 23:38:13 ERROR User login failed<br>host = Pradeepreddys-MacBook-Pro.local source = system_log_example_1000 (4).txt sourcetype = Analysis of logs                            |

Splunk Introduction - Analysis | Search | Splunk 9.3.1

localhost:8000/en-GB/app/search/search?q=search%20index%3Dsystem\_logs\_1\*ERROR\*%20%0A%27Ctop%20source&earliest=0&latest=&display.page.search.mode=... All time

splunk>enterprise Apps Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards

### New Search

Save As Create Table View Close

1 index=system\_logs\_1\*ERROR\*  
2 | top source All time

✓ 328 events (before 09/04/2025 18:58:51.000) No Event Sampling Job Smart Mode

Events Patterns Statistics (1) Visualization

20 Per Page Format Preview

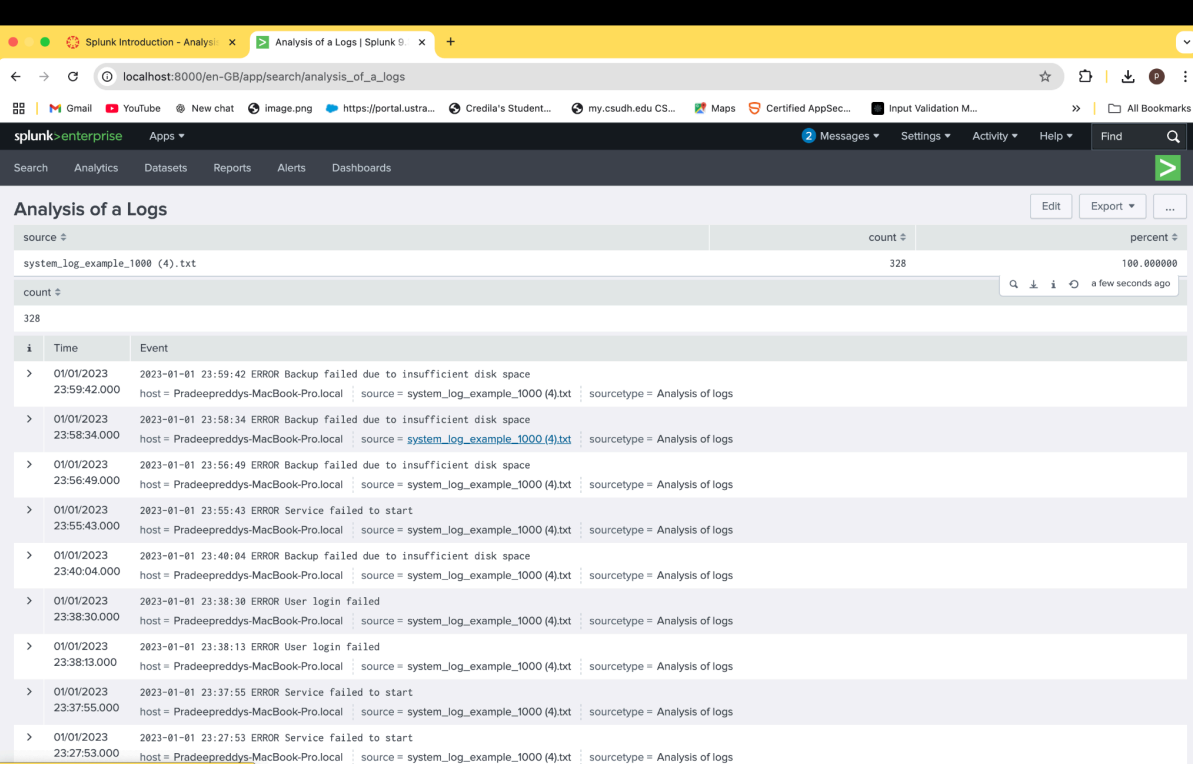
| source                          | count | percent    |
|---------------------------------|-------|------------|
| system_log_example_1000 (4).txt | 328   | 100.000000 |

## Dashboard Creation

Several search queries were saved as dashboard panels. A custom dashboard named **System Log Dashboard** was created, containing:

- Total count of ERROR logs.
- Timechart of log events over time.
- Breakdown of logs by severity (ERROR, INFO, WARN).
- Top log sources (if applicable).

This dashboard provides a clear, interactive view of the system's log activity for monitoring and analysis.



The screenshot displays the Splunk web interface for a dashboard titled "Analysis of a Logs". The search bar shows the query "system\_log\_example\_1000 (4).txt" with 328 results. The results are displayed in a table with columns for Time and Event. The events are all ERROR messages from "Pradeepreddys-MacBook-Pro.local".

| i | Time                    | Event                                                                                                                                                                                       |
|---|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 01/01/2023 23:59:42.000 | 2023-01-01 23:59:42 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs |
| > | 01/01/2023 23:58:34.000 | 2023-01-01 23:58:34 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs |
| > | 01/01/2023 23:56:49.000 | 2023-01-01 23:56:49 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs |
| > | 01/01/2023 23:55:43.000 | 2023-01-01 23:55:43 ERROR Service failed to start<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                      |
| > | 01/01/2023 23:40:04.000 | 2023-01-01 23:40:04 ERROR Backup failed due to insufficient disk space<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs |
| > | 01/01/2023 23:38:30.000 | 2023-01-01 23:38:30 ERROR User login failed<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                            |
| > | 01/01/2023 23:38:13.000 | 2023-01-01 23:38:13 ERROR User login failed<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                            |
| > | 01/01/2023 23:37:55.000 | 2023-01-01 23:37:55 ERROR Service failed to start<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                      |
| > | 01/01/2023 23:27:53.000 | 2023-01-01 23:27:53 ERROR Service failed to start<br>host = Pradeepreddys-MacBook-Pro.local   source = system_log_example_1000 (4).txt   sourcetype = Analysis of logs                      |